Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: A Denial-of-Service (DoS) attack known as a TCP SYN flood attack. The attacker sends a large number of TCP SYN requests to the web server, overwhelming its resources and preventing legitimate users from accessing the website.

The logs show that: A large number of TCP SYN packets are coming from an unknown IP address. The web server is receiving more connection requests than it can handle, causing it to become overloaded and fail to respond to legitimate requests.

This event could be: A TCP SYN flood attack, which is a type of Denial-of-Service (DoS) attack. The attacker may also use IP spoofing to change the source IP addresses and bypass IP-based firewall restrictions. This attack makes the website unavailable to employees and other legitimate visitors.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:
1. **SYN:** The client sends a SYN packet to the web server to request a connection.

2. **SYN-ACK:** The server responds with a SYN-ACK packet to acknowledge the request and indicate that it is ready to establish the connection.

3. **ACK:** The client sends an ACK packet to confirm the response, completing the three-way handshake and establishing the TCP connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: A malicious actor sends a large number of SYN packets to the web server, initiating many connection requests. The server responds with SYN-ACK packets and reserves resources while waiting for the final ACK packets. In a SYN flood attack, the attacker does not complete these handshakes, leaving many connections half-open. As these incomplete connections accumulate, the server's connection queue and resources become exhausted, preventing legitimate users from establishing connections.

Explain what the logs indicate and how that affects the server: The logs indicate an unusually high volume of TCP SYN packets originating from an unknown IP address. This traffic pattern is consistent with a TCP SYN flood attack. The large number of incomplete connection requests overwhelms the web server, causing slow response times and connection timeouts. As a result, employees cannot access the company's sales website to research vacation packages for clients.

The attack can negatively affect the organization by disrupting business operations, reducing employee productivity, damaging customer trust, and potentially causing financial losses. To prevent future attacks, the organization should implement SYN flood protection, configure firewall and intrusion prevention rules, use SYN cookies and connection rate limiting, and consider a DDoS protection service. Blocking a single IP address may not be sufficient because attackers can use IP spoofing or multiple source addresses to continue the attack.